

A.7 Mock Exam:

- Hot Dark Matter: Relativistic (neutrinos) ($kT \gg mc^2$)
- Warm Dark Matter: $kT \sim mc^2$
- Cold Dark Matter: $kT \ll mc^2$ (Supersymmetry, WIMPs, etc...)

* Power law Spectra: $\Delta^2(k)$ (Quantum "Box")

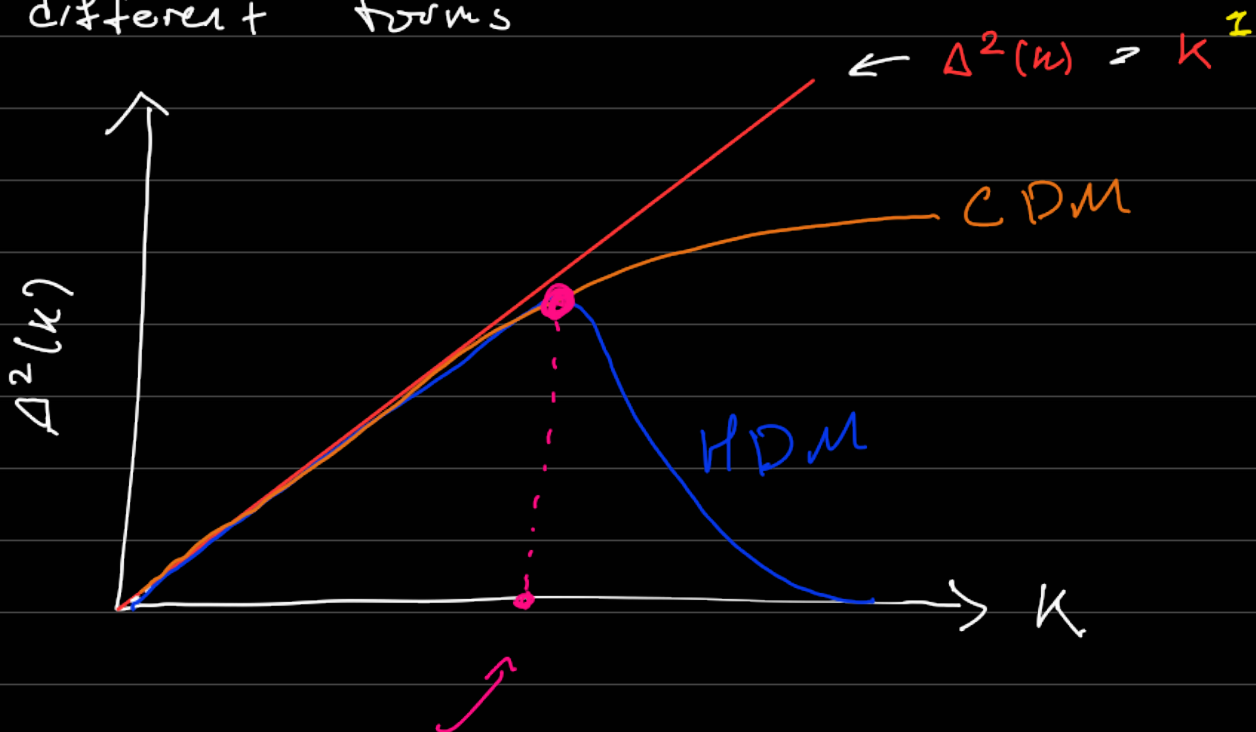
• Defined as

$$\Delta^2(k) \equiv \left(\frac{\text{Vol}}{2\pi^2} \right) 4\pi k^3 \underbrace{P(k)}_{\text{Power spectrum}}$$

$\Rightarrow$ Assume

$$\Delta^2(k) \propto k^n$$

- Both cold & dark matter start as $n=1$ but after the peak they have different forms



Peak occurs when $\rho_r = \rho_m$

ISM Problems:

- Ionization: $[S_*] \sim s^{-1}$ (# of ionizing photons per unit time)
- Recomb:

$$R_1 = \alpha_{\text{rec}} n_e \quad \left(\text{Recombination rate for a single electron} \right)$$

$$[R_2] \sim s^{-1}$$

→ Total Recomb is then

$$dR_{\text{tot}} = N_H R_1 = N_H \alpha_{\text{rec}} n_e$$

$$N_H = (4\pi R^2 dR) n_H$$

so the total:

$$\begin{aligned} dR_{\text{tot}} &= (4\pi R^2 dR n_H) \alpha_{\text{rec}} n_e \\ &= 4\pi R^2 dR n_H^2 \alpha_{\text{rec}} \end{aligned}$$

$$n_H = n_e = n_p$$

$$R_{\text{tot}} = \int 4\pi R^2 n_H^2 \alpha_{\text{rec}} dR$$

→ Equilibrium occurs: $R_{\text{tot}} = S_*$

Tully-Fisher: Spirals $\Rightarrow I = I_0 e^{-R/R_H}$

(i) Total Luminosity:

$$L_{\text{tot}} = \int_0^{2\pi} d\phi \int_0^\infty dR R I_0 e^{-R/R_H}$$

$\Rightarrow$

$$L_{\text{tot}} = 2\pi I_0 R_H^2$$

(ii) Virial theorem: $2K + U = 0$

$$2\left(\frac{1}{2} M V^2\right) - \alpha \frac{GM^2}{R} = 0$$

$$\Rightarrow V^2 = \alpha \frac{GM}{R}$$

(ii) Divide by Luminosity L to get

$$\mu = \left[\frac{M}{L} \right] \quad \text{on the RHS}$$

$$V^2 = \alpha \underline{G} \left[\underline{M} \right] = \underline{\alpha G} \mu$$

(iii) Assume μ is a constant

(iv) Assume L is given by total
 $L = 2\pi I_0 R^2$

(v) Replace R $\uparrow$ above

$$R = \left(\frac{L}{2\pi I_0} \right)^{1/2}$$

$$\rightarrow \frac{V^2}{L} = \frac{\alpha G M}{R} = \frac{\alpha G M (2\pi I_0)^{1/2}}{L^{1/2}}$$

$$V^2 = L^{1/2} \alpha G M (2\pi I_0)^{1/2}$$

(vi) Square

$$V^4 = L \left((\alpha G)^2 2\pi \right) I_0 \mu^2$$

$$\therefore L = \underbrace{\left(\frac{V^4}{I_0 \mu^2} \right) \left(\frac{1}{(\alpha G)^2 2\pi} \right)}_{\text{constant}}$$

Black Holes: $R_{\text{sch}} = \frac{2GM_{\text{BH}}}{c^2}$

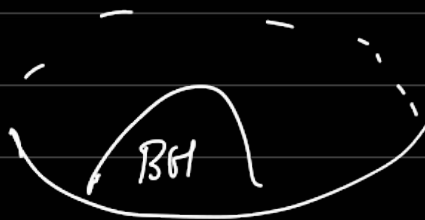
• Assume energy comes from gravity:

$$E = \epsilon (mc^2)$$

$\uparrow$ radiative eff. $\rightarrow$ rest mass

$$\bullet E_{\text{in}} = L = E_{\text{out}} = -(\dot{E})$$

Eddington limit: $F_g = (F_r)_e$



$$F_g = \frac{G M_{BH} (m_p + m_e)}{R^2}$$

$m_p + m_e \approx m_p$

$$F_g = \frac{G M_{BH} m_p}{R^2}$$

Both $m_p + m_e$ since the plasma is "frozen".

$$(F_r)_{tot} = \frac{dp}{dt}, \quad E = pc \text{ (photons)}$$

$$= \frac{d}{dt} \left(\frac{E}{c} \right) = \frac{1}{c} \frac{dE}{dt}$$

Luminosity

$$(F_r)_{tot} = \frac{1}{c} L$$

→ we only want the force on the cross-section of the electron so

$$(F_r)_e = \left(\frac{\sigma_T}{4\pi R^2} \right) (F_{tot}) = \frac{\sigma_T}{4\pi R^2} \left(\frac{L}{c} \right)$$

• set equal: $\frac{G M_{BH} m_p}{R^2} = \frac{\sigma_T}{4\pi R^2} \frac{L}{c}$

$$\therefore L_{edd} = \left(\frac{4\pi c G m_p}{\sigma_T} \right) M_{BH}$$

* Mass of B.H.: $L_{edd} = \frac{dE}{dt} = \frac{d}{dt} (E m c^2)$

$$= \dot{m} c^2$$

• Relate mass to mass of B.H.

$$\dot{M}_{BH} = (1-\epsilon) \dot{m}$$

• Replace:

$$\frac{Gc^2}{(1-\epsilon)} \dot{M}_{BH} = \left(\frac{4\pi c G m_p}{\sigma_T} \right) M_{BH}$$

define:

$$K \equiv \left((1-\epsilon) \frac{4\pi c G m_p}{\sigma_T G c} \right)$$

• then

$$\dot{M}_{BH} = K M_{BH}$$

$$\text{or } \frac{dM_{BH}}{M_{BH}} = K dt$$

• Integrate: $\int \frac{dM_{BH}}{M_{BH}} = \int K dt$

$$\ln(M_{BH}) = Kt + C$$

$$M_{BH} = \underbrace{e^C}_{\text{call } M_0} e^{Kt} \quad (\text{constant})$$

$$M_{BH} = M_0 e^{Kt}$$

Good luck!



